

Roll No.....

MID Semester Examination 2025 (B.Arch)

Name of the Program: B. Tech (All Branches)

Semester: 2nd

Name of the Course: Engineering Mathematics -II

Course Code: TMA 201

Time: 1:30 Hour

Maximum Marks: 50

Note: -

- (i) Answer all the questions by choosing any one of the sub questions.
- (ii) Each question carries 10 marks.

Q1 (10 marks)		CO1
a)	Define Exact Differential equation. Also state the necessary and sufficient condition that an equation of first order and first degree be exact. Solve $\left(y + \frac{1}{3}y^2 + \frac{1}{2}x^2\right)dx + \frac{1}{4}(1 + y^2)x dy = 0$	
OR		
b)	Find the complete solution of the following differential equation $(D - 2)^2 y = 8(e^{2x} + \sin 2x + x^2)$	
Q2 (10 marks)		CO1
a)	Find the complete solution of the following differential equation $(x^2 D^2 + xD + 1)y = \log x \sin(\log x)$	
OR		
b)	Find the complete solution of the following differential equation by the method of variation of parameters: $(D^2 - 2D)y = e^x \sin x$	
Q3 (10 marks)		CO1
a)	Solve the following differential equation $(D^2 - 2D + 1)y = x e^x \cos x;$	
OR		
b)	Find the Laplace transform of the following function $f(t) = \frac{e^{-at} - e^{-bt}}{t}$.	
Q4 (10 marks)		CO2
a)	Express the following function in terms of unit step function and find its Laplace transform $f(t) = \begin{cases} t^2, & 0 < t \leq 2 \\ 4t & t > 2 \end{cases}$	
OR		

b)	Find the Laplace transform of the following periodic function (Rectified sine wave of Period a) $f(t) = \sin\left(\frac{\pi t}{a}\right)$ for $0 < t < a$.	
Q5	(10 marks)	
a)	State Convolution theorem and hence evaluate $L^{-1}\left\{\frac{p}{(p^2+1)(p^2+4)}\right\}$.	
	OR	
b)	Using Laplace transformation, solve the following differential equation $(D^2 + 9)y = \cos 2x$, if $y(0) = 1, y\left(\frac{\pi}{2}\right) = -1$.	CO2